

Supplementary Information

Single-Cell Profiling Decodes Patagium Development in Gliding Mammal

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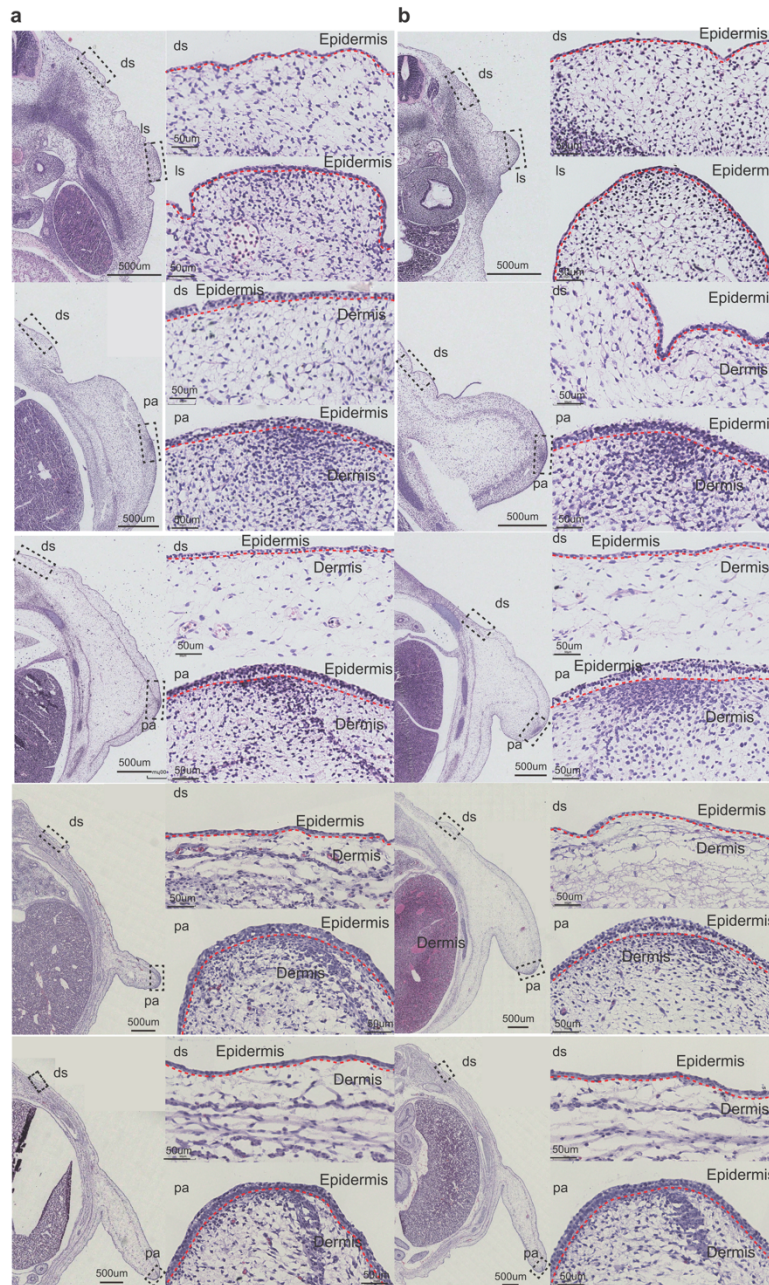
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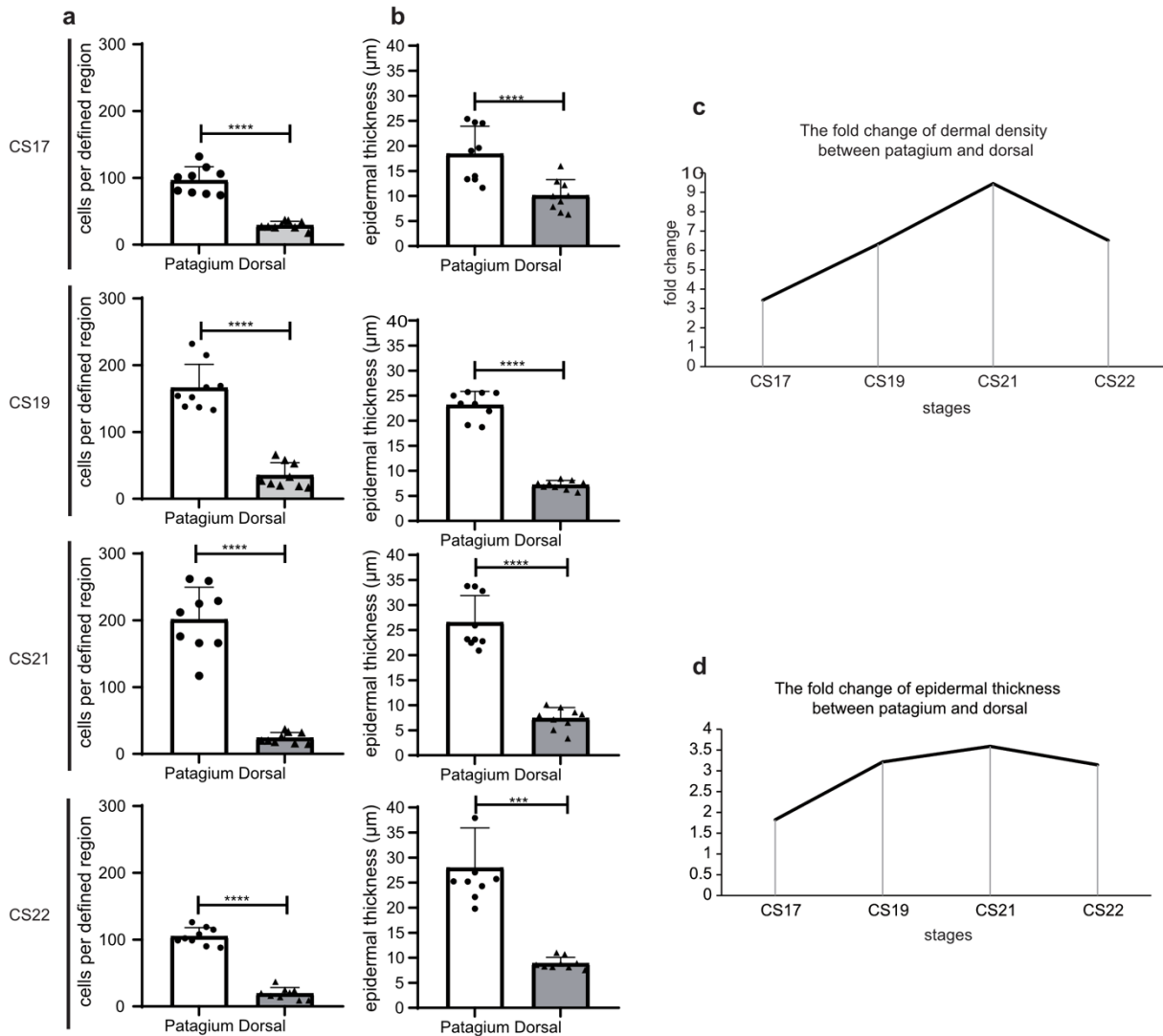
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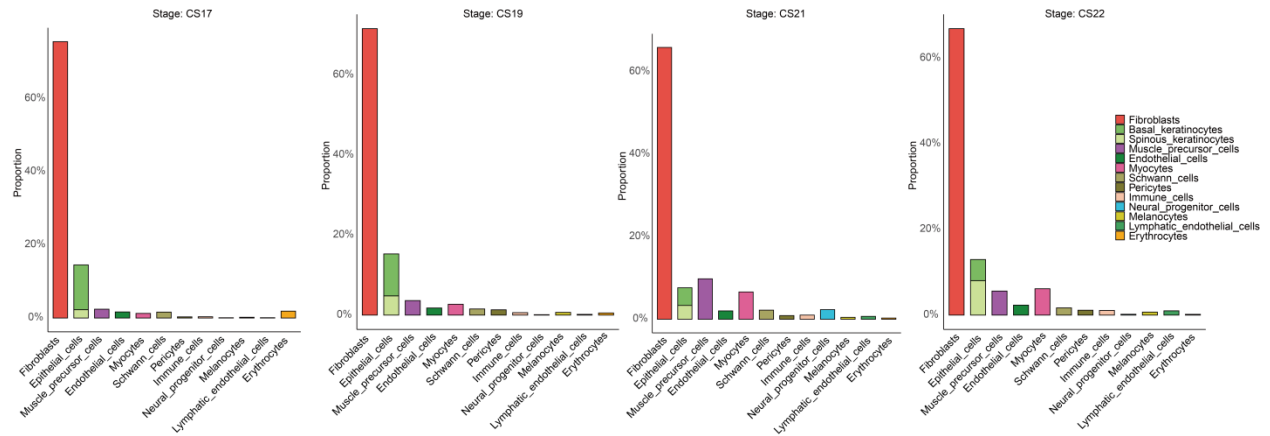
Supplementary Figures



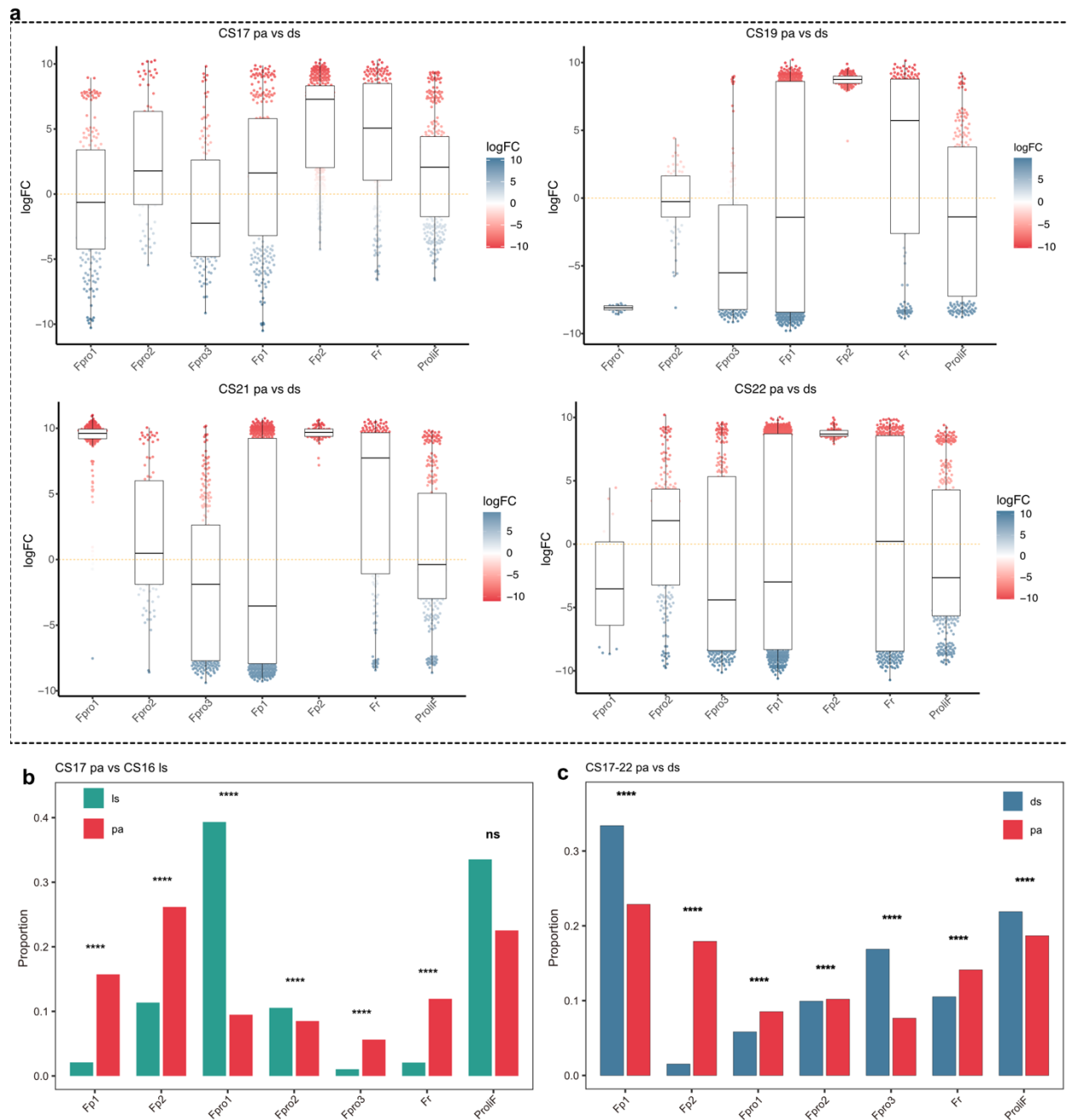
Supplementary Figure 1: Histological analysis of embryonic dorsal skin, lateral skin and patagium in *Troglodytes xanthipes*. Representative sections from three independent embryos showed similar histological features. In **a** and **b**, left panels show low-magnification views (4×), and right panels show corresponding high-magnification views (20×). The red dashed lines indicate the basement membrane, demarcating the epidermis and dermis. ds, dorsal skin; ls, lateral skin; pa, patagium.



Supplementary Figure 2: Quantification of dermal cell density and epidermal thickness during embryonic development in *T. xanthipes*. **a** Quantification of dermal cell density in the dermal layer of patagium and dorsal skin. **b** Quantification of epidermal thickness in patagium and dorsal skin. **c** Fold change in dermal cell density between patagium and dorsal skin. **d** Fold change in epidermal thickness between patagium and dorsal skin. Data are presented as mean $\pm$ s.d. from three embryos per stage, with three sections quantified per embryo (nine measurements per group are shown). Statistical significance was assessed using two-tailed unpaired t-tests. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.001$; **** $P < 0.0001$. Source data are provided as a Source Data file.

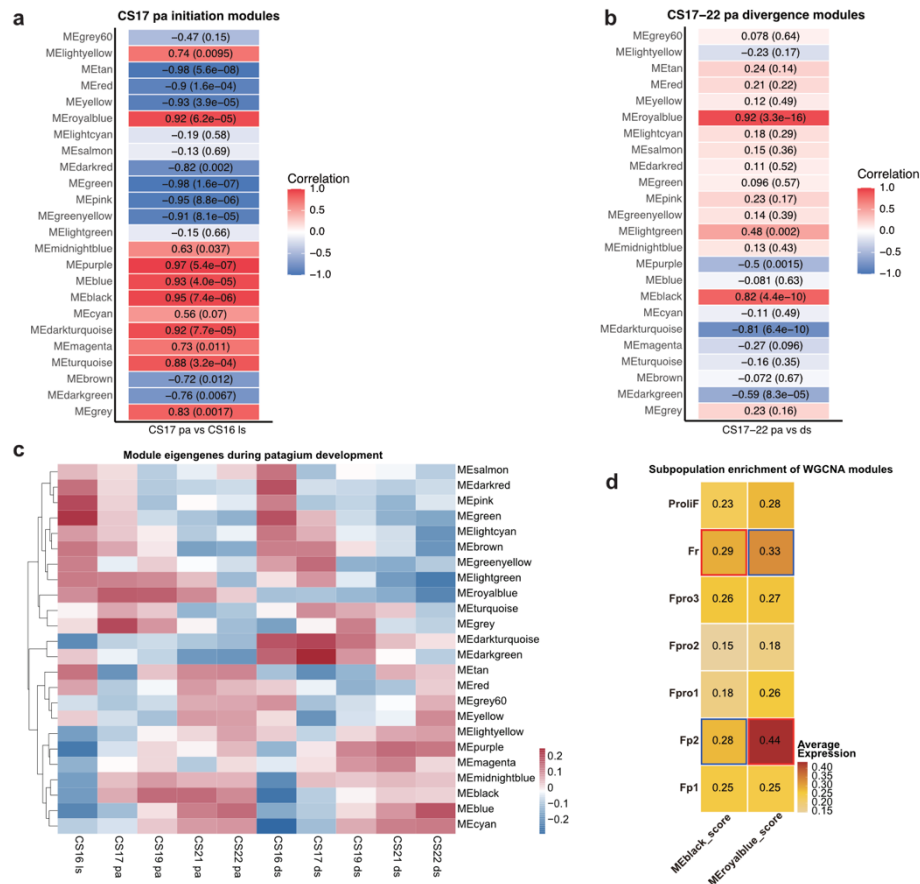


Supplementary Figure 3: Proportions of the cell types during embryonic patagium development. Proportions of the 13 major cell types (Supplementary Table 4) across developmental stages (CS17, CS19, CS21 and CS22) in the developing patagium.

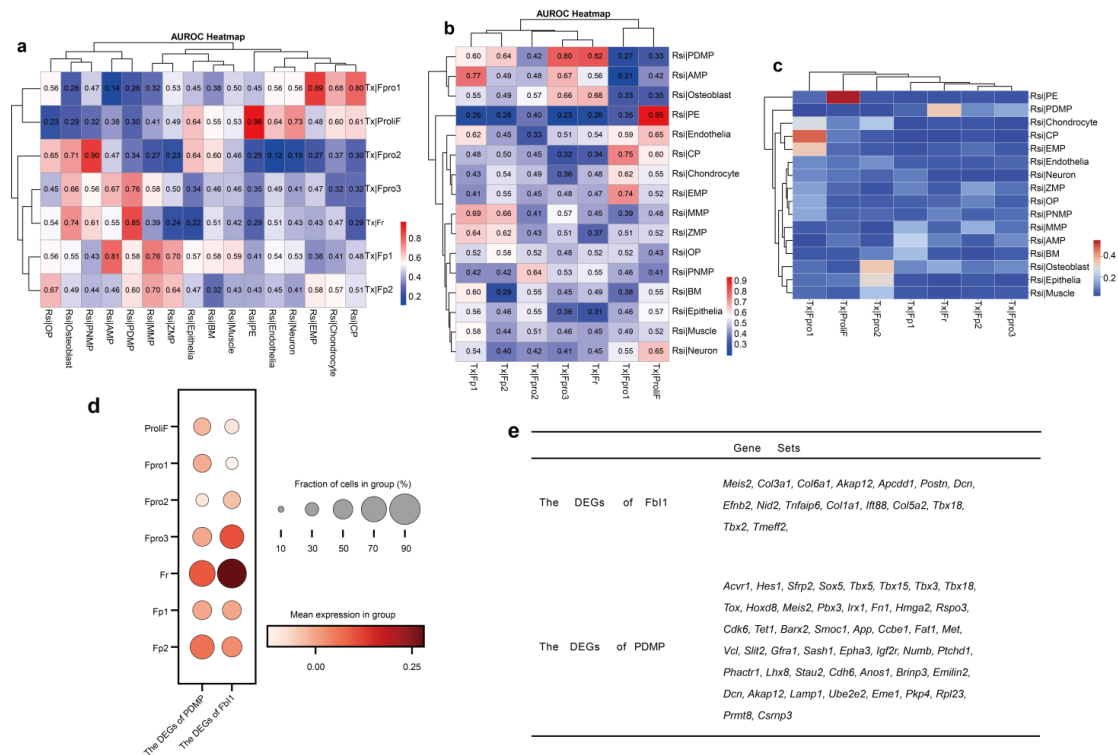


Supplementary Figure 4: Differential abundance and compositional dynamics of fibroblast subpopulations during patagium development. **a** Milo neighborhood differential abundance analysis of fibroblast subpopulations between patagium and dorsal skin at CS17, CS19, CS21 and CS22. Points are coloured according to logFC values, with red indicating enrichment in patagium and blue indicating enrichment in dorsal skin. The y-axis shows logFC, Positive logFC values indicate enrichment in patagium, whereas negative values indicate enrichment in dorsal skin. Points are colored by logFC. Box plots show the distribution of neighborhood logFC values; centre

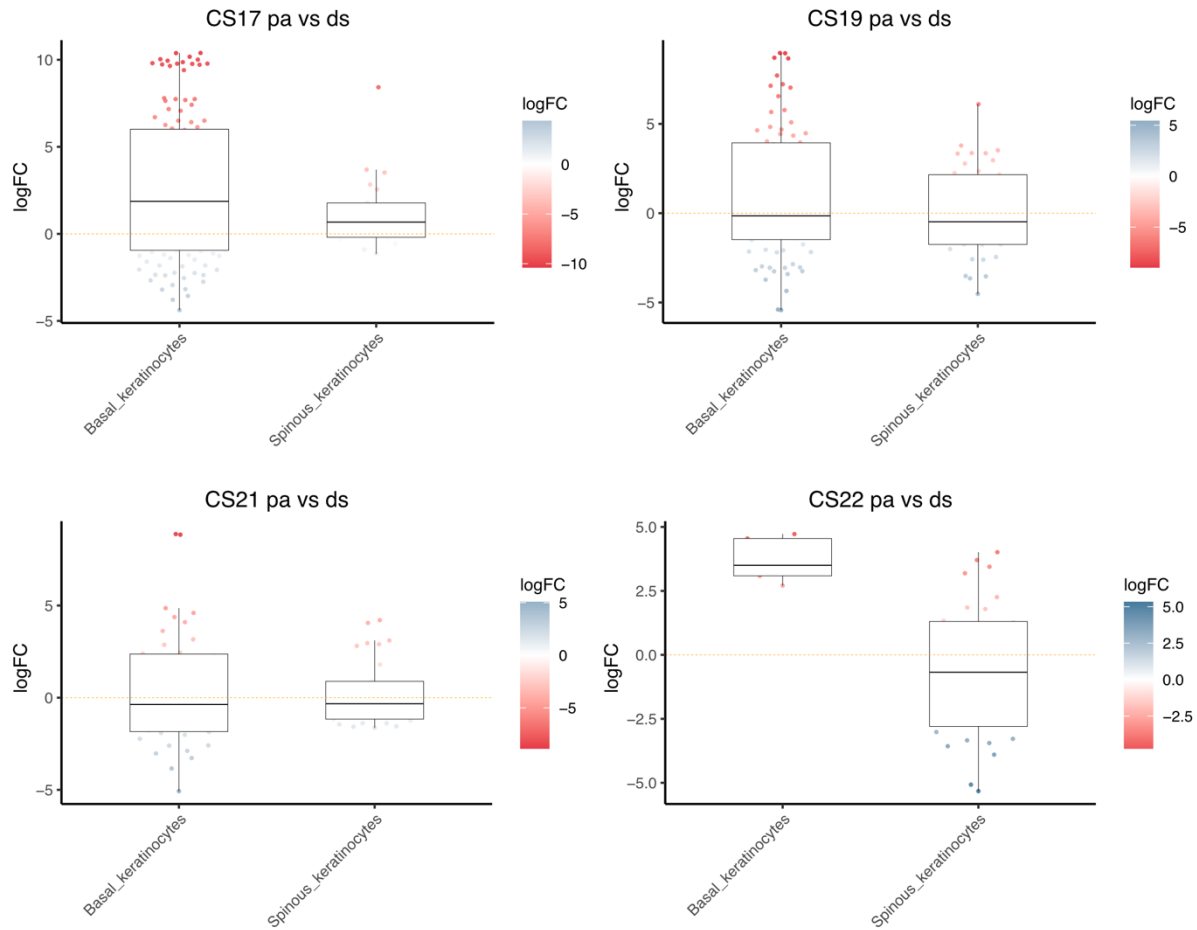
lines indicate medians, box limits indicate the first and third quartiles, and whiskers extend to the most extreme data points within 1.5 x the interquartile range. **b** Comparison of fibroblast subpopulation proportions between CS17 patagium and CS16 lateral skin. **c** Comparison of fibroblast subpopulation proportions between patagium and dorsal skin across CS17-CS22 developmental stages. Statistical significance in **b** and **c** was assessed using chi-square tests based on downsampled cell counts. Exact P values are provided in the Source Data file. ****P < 0.0001; ns, not significant. Source data are provided as a Source Data file.



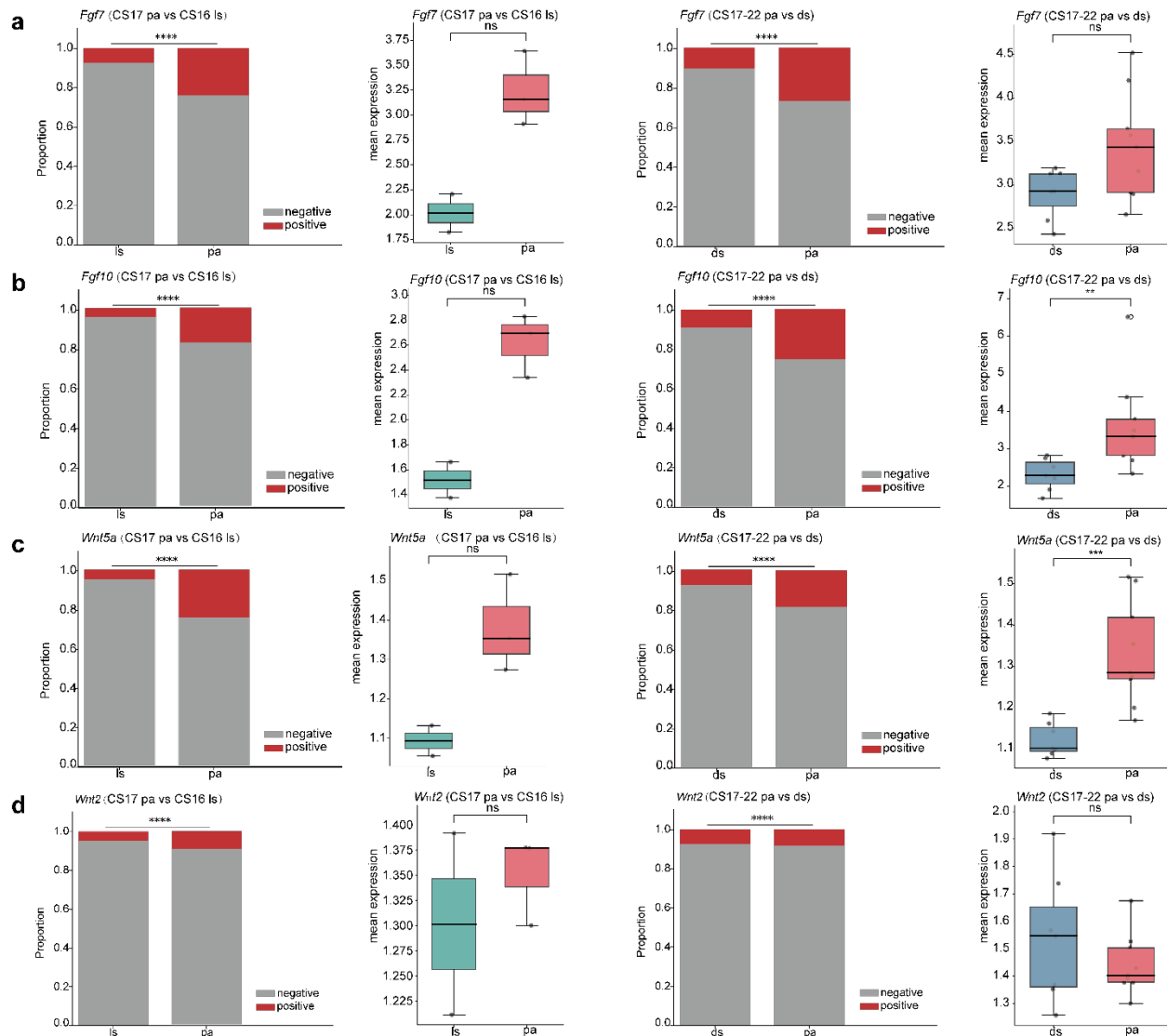
Supplementary Figure 5: WGCNA module-trait relationships across tissues and fibroblast subpopulations. **a,b** Module-trait relationship heatmaps. **a** Correlations between WGCNA modules and the CS17 patagium versus CS16 lateral skin comparison. **b** Correlations between WGCNA modules and the patagium versus dorsal skin comparison across CS17-CS22 stages. Rows represent WGCNA modules, and colour intensity indicates the strength and direction of module-trait correlations, with red indicating positive correlations and blue indicating negative correlations. Values show correlation coefficients, with corresponding P values in parentheses. **c** Module eigengene (ME) values across tissues during development. **d** Mean WGCNA module scores for selected modules across fibroblast subpopulations. Colour intensity indicates the mean module score. Red and blue outlines indicate the fibroblast subpopulations with the highest and second-highest module scores, respectively. Source data are provided as a Source Data file.



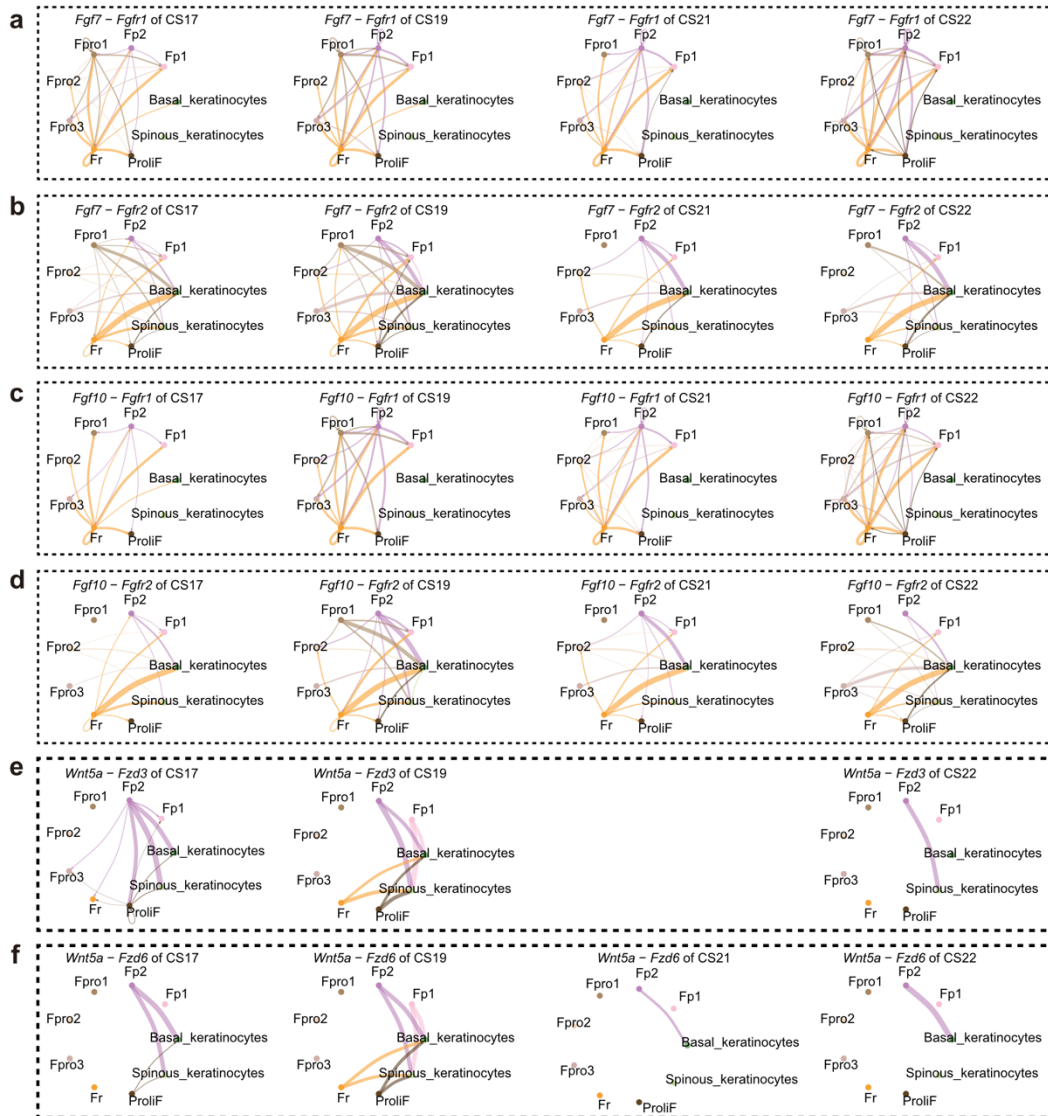
Supplementary Figure 6: Cross-species conservation of patagium cell populations between flying squirrel and bat. **a, b** MetaNeighbor analysis of transcriptional conservation of cell populations between flying squirrel (*Troglodytes xanthipes*) and bat (*Rhinolophus sinicus*) patagium. Heatmaps show area under the receiver operating characteristic (AUROC) values, reflecting the replicability of cell population transcriptional profiles across species. Higher AUROC values indicate stronger cross-species correspondence between cell types. **a** The model was trained on the *T. xanthipes* dataset and used to predict the corresponding cell populations in *R. sinicus*. **b** The model was trained on the *R. sinicus* dataset and used to predict corresponding cell populations in *T. xanthipes*. **c** SAMap cross-species alignment scores between *T. xanthipes* and *R. sinicus* cell populations. The alignment scores reflect the strength of cross-species cell-type alignment based on transcriptomic similarity. **d** Dot plots showing the expression patterns of differentially expressed gene (DEG) sets from two previously reported bat cell populations — *Pdgfr*⁺ mesenchymal progenitors (PDMPs) in the developing bat wing¹ and Fbl1 fibroblasts in the chiropatagium² — across flying squirrel fibroblast subpopulations. Dot size indicates the proportion of cells expressing each gene, and colour intensity represents the mean normalized expression level. **e** Gene lists corresponding to the two DEG sets used in d. Source data are provided as a Source Data file.



Supplementary Figure 7: Milo neighborhood differential abundance analysis of epithelial subpopulations during patagium development. Each point represents a Milo neighborhood assigned to the indicated epithelial subpopulation. The y-axis shows logFC, with positive values indicating enrichment in patagium and negative values indicating enrichment in dorsal skin. Points are colored by logFC. Box plots show the distribution of neighborhood logFC values, with red indicating enrichment in patagium and blue indicating enrichment in dorsal skin; centre lines indicate medians, box limits indicate the first and third quartiles, and whiskers extend to the most extreme data points within 1.5 x the interquartile range. Source data are provided as a Source Data file.

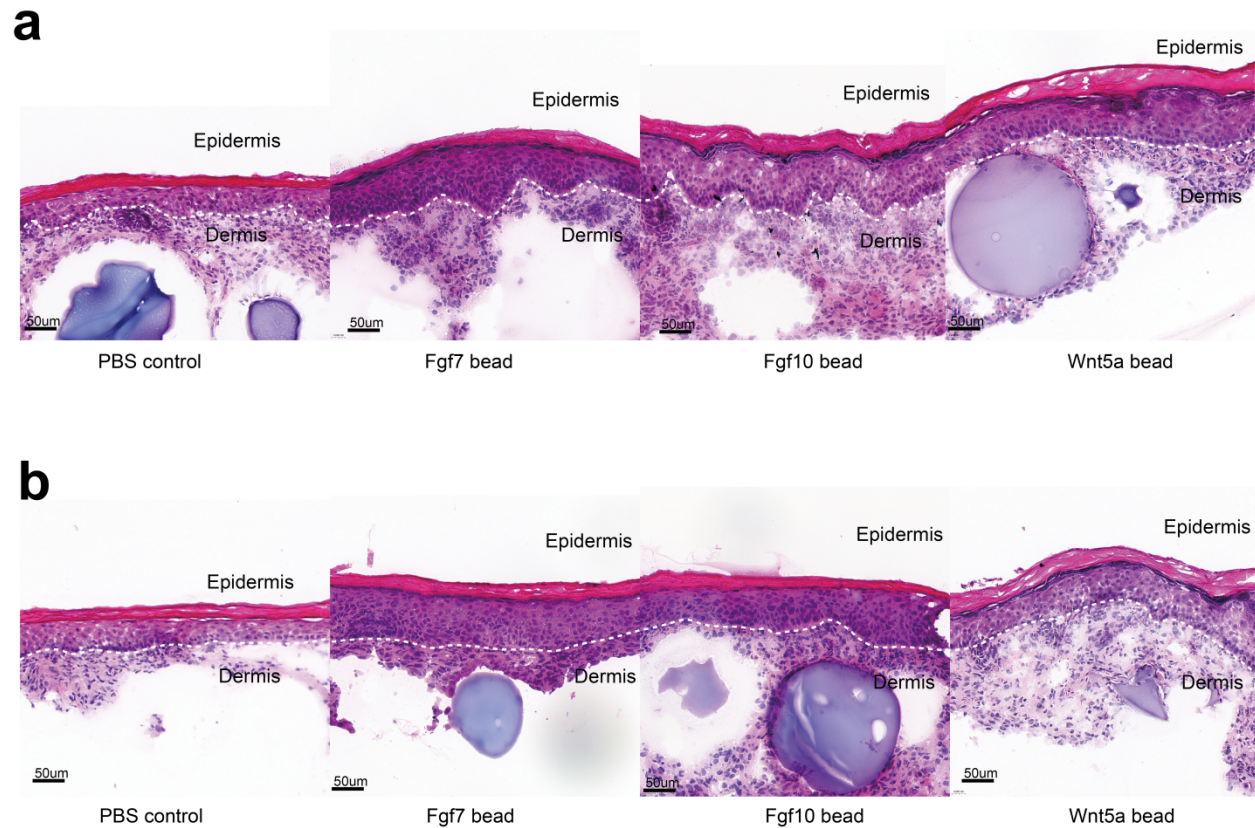


Supplementary Figure 8: Expression patterns of *Fgf7*, *Fgf10*, *Wnt5a* and *Wnt2*. For each gene, bar plots show the proportion of fibroblasts with detectable or undetectable expression of each ligand gene, and box plots show sample-level mean expression among ligand-positive cells only. Left and right panels correspond to CS17 patagium versus CS16 lateral skin and CS17-CS22 patagium versus dorsal skin comparisons, respectively. In box plots, each point represents one sample-level mean calculated from ligand-positive cells. Centre lines indicate medians, box limits indicate the first and third quartiles, and whiskers extend to the most extreme data points within $1.5 \times$ the interquartile range. Statistical significance was assessed using two-tailed Mann-Whitney U tests. * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$ and **** $P < 0.0001$; ns, not significant. Exact P values, sample sizes and underlying values are provided in the Source Data file. Source data are provided as a Source Data file.

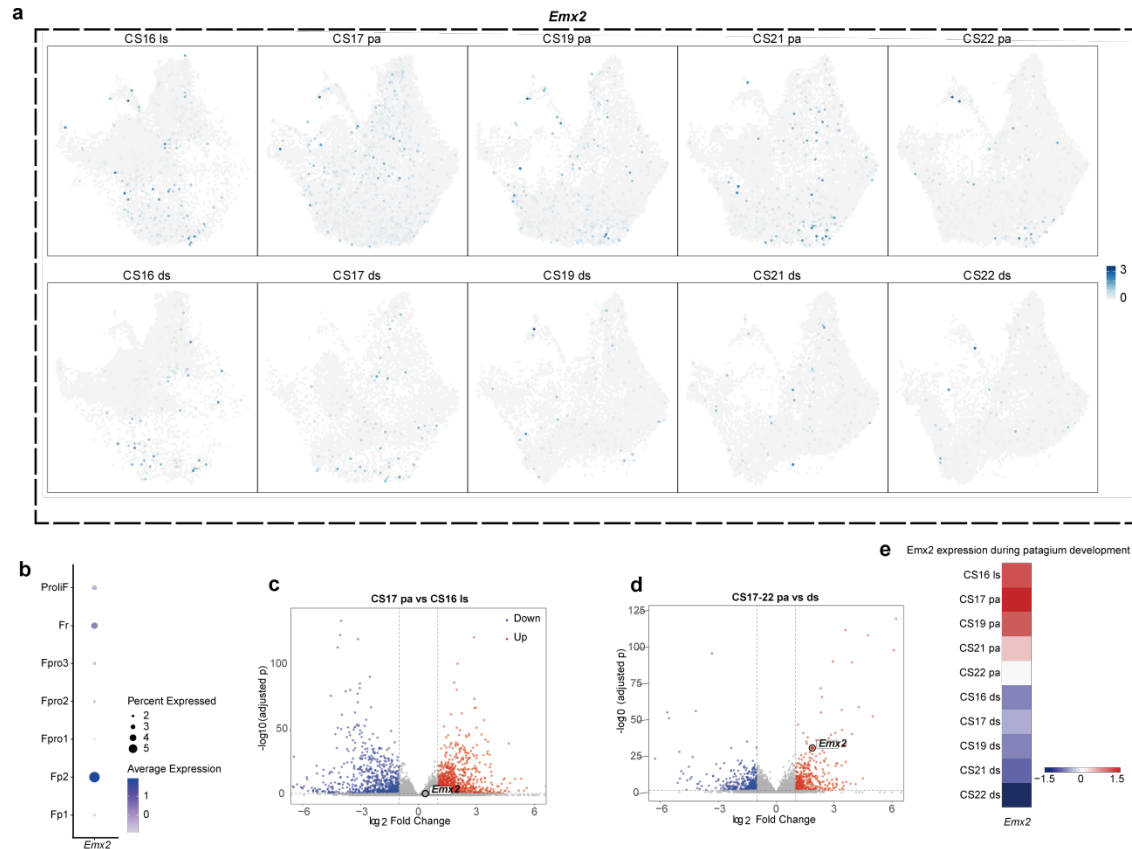


Supplementary Figure 9: Predicted FGF and non-canonical WNT signaling interactions between fibroblast and epithelial subpopulations in the patagium. **a** Circle plots showing inferred *Fgf7-Fgfr1* signaling interactions between fibroblast and epithelial subpopulations across developmental stages (CS17-CS22). **b** Circle plots showing inferred *Fgf10-Fgfr1* signaling interactions between fibroblast and epithelial subpopulations across developmental stages (CS17-CS22). **c** Circle plots showing inferred *Fgf7-Fgfr2* signaling interactions between fibroblast and epithelial subpopulations across developmental stages (CS17-CS22). **d** Circle plots showing inferred *Fgf10-Fgfr2* signaling interactions between fibroblast and epithelial subpopulations across developmental stages (CS17-CS22). **e** Circle plots showing inferred *Wnt5a-Fzd3* signaling interactions between fibroblast and epithelial subpopulations across developmental stages (CS17, CS19 and CS22). **f** Circle plots showing inferred *Wnt5a-Fzd6* signaling interactions between

fibroblast and epithelial subpopulations across developmental stages (CS17-CS22). Nodes represent cell subpopulations, node colours indicate fibroblast or epithelial subpopulation identities, arrows indicate the inferred direction of communication from sender to receiver populations, and edge width represents the inferred interaction strength. Source data are provided as a Source Data file.



Supplementary Figure 10: Histological analysis of embryonic mouse dorsal skin explants following bead implantation. Representative H&E-stained sections from three independent explant culture experiments showing consistent phenotypes. **a,b** Representative images at 20x magnification. The white dashed lines indicate the basement membrane, demarcating the epidermis (above) from the dermis (below).



Supplementary Figure 11: Temporal expression dynamics of *Emx2* across fibroblast subpopulations and tissues during developmental stages. **a** Feature plots showing the expression distribution of *Emx2* in fibroblast populations and tissues during development. Colour intensity (blue gradient) indicates normalized gene expression level (log-transformed counts). **b** Dot plot summarizing *Emx2* expression across fibroblast subpopulations. Dot size corresponds to the proportion of cells expressing *Emx2* in each subcluster, and colour intensity (purple gradient) represents the average normalized expression level. **c, d** Volcano plot of differentially expressed genes (DEGs) from the bulk RNA-seq analysis (CS17 patagium versus CS16 lateral skin and CS17-CS22 patagium versus dorsal skin comparison). Genes with an adjusted P value of less than 0.05 and $|\log_2 \text{ fold change}| > 1$ are highlighted (red: up-regulated; blue: down-regulated); *Emx2* is marked. **e** Heatmap showing the normalized expression of *Emx2* across tissues at distinct developmental stages. Colour intensity indicates standardized expression values (z-scores), with red indicating higher relative expression and blue indicating lower relative expression. Source data are provided as a Source Data file.

Supplementary Tables

Supplementary Table1. An overview of the assembly statistics.

Genomic information	<i>Trogopterus xanthipes</i>
Total length (Gbp)	913.83
Genome size (Gbp)	3.17
Number of scaffolds	6851
Total length of scaffolds (Gb)	3.18
N50 length of contigs (Mb)	4.085
N50 length of scaffolds (Mb)	192
GC content (%)	40.55
Core eukaryotic genes mapping approach (CEGMA) (%)	94.76
Benchmarking universal single-copy orthologs (BUSCO) (%)	98.40
Total repeat content (%)	34.35
Mapping rate of small fragments (%)	98.97
Number of protein-coding genes of annotation	23,706

Supplementary Table2. Genome annotation for *Trogopterus xanthipes*.

Database		Annotated number	Annotated percent (%)
Swiss-Prot		20,257	85.45
Tremble		20,313	85.69
KEGG		20,434	86.20
All		20,073	84.67
InterPro	Pfam	18,434	77.76
	GO	15,014	63.33
Annotated		20,478	86.38
Total		23,706	-

Supplementary Table3. Detailed information for the single-nucleus RNA sequencing samples.

Sample ID [#]	Stage	Tissue	Sample number [*]	Library type
CS16_ls1	CS16	Lateral skin	2	10X Genomics (v3)
CS16_ls2	CS16	Lateral skin	2	10X Genomics (v3)
CS17_pa1	CS17	Patagium	1	10X Genomics (v3)
CS17_pa2	CS17	Patagium	2	10X Genomics (v3)
CS17_pa3	CS17	Patagium	2	10X Genomics (v3)
CS19_pa1	CS19	Patagium	1	10X Genomics (v3)
CS19_pa2	CS19	Patagium	1	10X Genomics (v3)
CS21_pa1	CS21	Patagium	1	10X Genomics (v3)
CS21_pa2	CS21	Patagium	1	10X Genomics (v3)
CS22_pa1	CS22	Patagium	1	10X Genomics (v3)
CS22_pa2	CS22	Patagium	1	10X Genomics (v3)
CS16_ds1	CS16	Dorsal skin	2	10X Genomics (v3)
CS16_ds2	CS16	Dorsal skin	2	10X Genomics (v3)
CS17_ds1	CS17	Dorsal skin	2	10X Genomics (v3)
CS19_ds1	CS19	Dorsal skin	2	10X Genomics (v3)
CS19_ds2	CS19	Dorsal skin	1	10X Genomics (v3)
CS21_ds1	CS21	Dorsal skin	1	10X Genomics (v3)
CS21_ds2	CS21	Dorsal skin	1	10X Genomics (v3)
CS22_ds1	CS22	Dorsal skin	1	10X Genomics (v3)
CS22_ds2	CS22	Dorsal skin	1	10X Genomics (v3)

Note: #: The sample ID corresponds to a sequencing library generated from a given tissue and developmental stage. *: The number of individual embryos contributing to each library. A value of 1 denotes that the library was generated from a single embryo; A value of 2 indicates that the library was generated from pooled tissues derived from two embryos.

Supplementary Table4. Annotated cell types and cell numbers together with their reference gene markers for annotation.

Cell type	Cell number	Markers used for cell type identification	References
Fibroblasts	142,561	<i>Colla1, Fbn1, Pdgfra, Fap, Prrx1</i>	3, 4, 5, 6
Basal keratinocytes	13,499	<i>Itga6, Tp63, Cxcl14</i>	7, 8
Spinous keratinocytes	8,066	<i>Dsg1, Cdh1, Spink5</i>	7
Muscle precursor cells	11,535	<i>Eya1, Eya2, Met, Pax7</i>	9
Endothelial cells	5,169	<i>Pecam1, Kdr, Cdh5, Ptprb, Emcn</i>	4, 10, 11
Myocytes	9,773	<i>Actc1, Tnnt1, Myh3</i>	11, 12
Neural progenitor cells	5,751	<i>Tox3, Nrxa3</i>	13, 14
Schwann cells	3,779	<i>Cdh19, L1cam, Erbb3</i>	15
Pericytes	2,116	<i>Igfbp7, Trpc6, Gucyl1a2</i>	16, 17, 18
Immune cells	2,028	<i>Ptprc</i>	10
Erythroblasts	1,599	<i>Epb42, Rhag, Tfrc</i>	19, 20, 21
Melanocytes	1,022	<i>Kit, Tyr, Fmn1</i>	22
Lymphatic endothelial cells	1,698	<i>Prox1</i>	23

Supplementary Table5. The key embryonic features of *T. xanthipes*.

Carnegie stage	Key embryonic features
CS16	fbv: The forebrain vesicle is clear. rp: The retinal pigment in the optic vesicle becomes well defined. fl: The forelimbs are seen to be divided into two regions, the proximal part consisting of the future limb girdle and arm, and the more peripheral part, which forms a circular or paddle-shaped hand plate (anterior foot plate). hl: The thigh, leg, and foot are becoming distinguishable, but there is not a circular footplate.
CS17	fbv: The forebrain vesicle develops. rp: It is possible to see the retinal pigment of the pigmented layer through the transparent cornea. fl: The hand plate (anterior foot plate) is no longer circular but exhibits definite digital rays. hl: The posterior foot plate has acquired a rounded digital plate, and is also distinguishable from the lower part of the leg. pa: The patagium primordium first appears on the lateral plane of the embryo as a ridge between the forelimb and the hindlimb.
CS19	pi: The pinna rapidly develops and forms a crest at right angles to the head. lv: lens vesicle is visible. fl: The wrists are now identifiable, and interdigital notches (idn) are to be seen separating the digits on the hand plate. hl: The digits on the foot plates are distinguishable. pa: The patagium extends distally and appears as an arc when viewed sideways.
CS21	pi: The growing pinna is turned forward and covers about one-half of the external auditory meatus. fl: Individual fingers are visible in the anterior foot plate, and the elbow (e) region is visible. hl: The interdigital notches (idn) on the foot plates are not yet separated.

CS22

pa: The patagium continues to extend and the arc increases.

el: The eyelids form across the eyes, which are open.

fl: Fingers separate.

hl: Toes separate, the heels (h) on the lower limb bud.

pa: The patagium continues to extend with the front and back limbs.

Supplementary Table6. Statistical analysis of the genome sequencing data for *Trogopterus xanthipes*.

Double terminal sequencing library	Inserts	Total data (Gb)	Fragment read length (bp)	Sequence coverage (X)
Illumina reads	350	420.59	150	116.8
PacBio reads	-	254.96	-	71
10X Genomics	-	238.28	-	66.2
Total	-	913.83	-	254

Supplementary Table7. The PacBio long read and Illumina short read RNA-sequencing transcriptome from multiple organs.

Tissue	RNA-seq data size (Gb)	Sequencing technology
Kidney	28.99	PacBio long read sequencing
Heart	14.72	PacBio long read sequencing
Brain	27.62	PacBio long read sequencing
Skin	12.33	PacBio long read sequencing
Blood	9.37	PacBio long read sequencing
Muscle	25.46	PacBio long read sequencing
Lung	27.68	PacBio long read sequencing
Liver	24.73	PacBio long read sequencing
Patagium	19.2	Illumina short read sequencing
Muscle	18.74	Illumina short read sequencing
Skin	16.07	Illumina short read sequencing
Blood	6.38	Illumina short read sequencing
Pituitary	18.7	Illumina short read sequencing
Brain	17.3	Illumina short read sequencing
Lung	18.5	Illumina short read sequencing
Pons	10.4	Illumina short read sequencing
Cerebellum	18.9	Illumina short read sequencing
Heart	14.8	Illumina short read sequencing
Eye	18.48	Illumina short read sequencing
Medulla oblongata	17.1	Illumina short read sequencing

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